

Simple categorical analysis (using R)

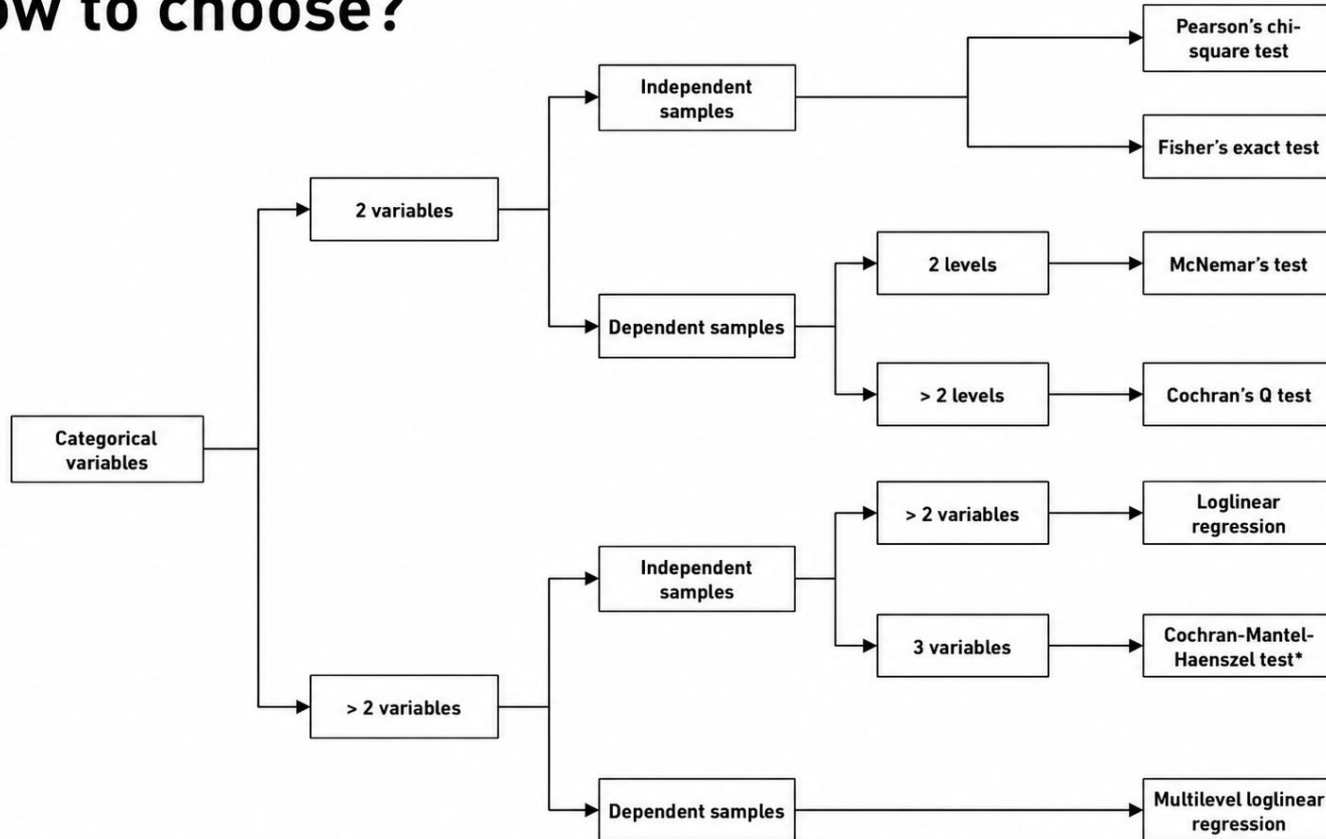
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How to choose?



Related concepts

1. Association

- Describe the relationship between 2 variables
- Measure of association (strength):
 - i. Relative risk
 - ii. Odds ratio

Relative risk

- Also known as risk ratio (RR)
- RR
 - Ratio of risk of an event in one group (i.e. exposed group) vs. the risk of the event in the other group (i.e. non-exposed group)
 - Measure the association between an exposure and an outcome
- Interpretation:
 - $RR=1$ Exposure does not affect the risk of outcome
 - $RR>1$ Exposure associated with higher risk of outcome
 - $RR<1$ Exposure associated with lower risk of outcome

Odds ratio (OR)

- Ratio of odds of an event in one group vs. the odds of the event in the other group.
- Measure the association between an exposure and an outcome
- Odds
 - likelihood of an event to occur
 - Odds = $P / (1-P)$, P is probability of an event to occur
- Interpretation:
 - OR=1 Exposure does not affect odds of outcome
 - OR>1 Exposure associated with higher odds of outcome
 - OR<1 Exposure associated with lower odds of outcome

OR vs RR

- $OR \approx RR$ only when the outcome is rare ($<10\%$)
- When, the outcome is relatively common, usually $OR > RR$ - thus, overestimate the true association between variables
- OR can be calculated for all (if not, most of) study designs, thus, commonly used
- But many against the use of OR in cohort and clinical trial studies as RR is better and can be calculated

- Odds $\neq$ risk (probability)
 - A RR of 3 means the risk of an outcome is increased threefold
 - A RR of 0.5 means the risk is cut in half
 - But an OR of 3 doesn't mean the risk is threefold; rather the odds is threefold greater
 - Interpretation of an OR must be in terms of odds, not probability

	Disease (case)	No disease (control)
Exposed	A	B
Not exposed	C	D

$$RR = \frac{\text{Disease risk of exposed } [A/(A+B)]}{\text{Disease risk of non-exposed } [C/(C+D)]}$$

$$OR = \frac{\text{Odds that a case was exposed } (A/C)}{\text{Odds that a control was exposed } (B/D)} = \frac{AD}{BC}$$

\$data			
	Lung_cancer		
Smoking	cancer	noncancer	Total
smoker	39	12	51
nonsmoker	27	28	55
Total	66	40	106

\$measure			
	odds ratio with 95% C.I.		
Smoking	estimate	lower	upper
smoker	1.00000	NA	NA
nonsmoker	3.37037	1.461559	7.772108

\$p.value			
	two-sided		
Smoking	midp.exact	fisher.exact	chi.square
smoker	NA	NA	NA
nonsmoker	0.004041203	0.00490976	0.003664719

OR:

- Smokers have 3.37 higher odds of getting a lung cancer than non-smokers
- OR is significant as 95% CI does not cross 1

\$data			
Smoking	Lung_cancer		Total
	nonsmoker	cancer	
nonsmoker	28	27	55
smoker	12	39	51
Total	40	66	106

\$measure			
Smoking	risk ratio with 95% C.I.		
	estimate	lower	upper
nonsmoker	1.000000	NA	NA
smoker	1.557734	1.143424	2.122166

\$p.value			
Smoking	two-sided		
	midp.exact	fisher.exact	chi.square
nonsmoker	NA	NA	NA
smoker	0.004041203	0.00490976	0.003664719

RR:

- Smokers have 1.56 higher risk (or 56% higher risk) of getting a lung cancer than non-smokers
- All 95% CI are significant as it does not cross 1

Simple categorical analysis

- Pearson's chi-square test
- Fisher's exact test
- McNemar's test
- Cochran's Q test
- Cochran-Mantel-Haenszel test

Pearson's chi-square test

- Aim: to assess the association between two categorical variables
- Condition:
 - Independent samples
 - Variables are categorical
- Example:
 - To determine the association between smoking and heart attack
- Assumptions:
 - Expected count < 5 :
 - $>20\%$ - Fisher's exact test
 - $<20\%$ - Pearson's chi-square test

- Presenting and interpreting the result:
 - In sentence: there is a significant association between smoking and lung cancer ($P = 0.004$)
 - In table:

Variable		Lung cancer n(%)	Non-lung cancer n(%)	n	X ² -statistics (df)	P-value*
Smoking	Yes	39 (76.5)	12 (23.5)	55	8.4 (1)	0.004
	No	27 (49.1)	28 (50.9)	51		

*Pearson's chi-square test

Fisher's exact test

- Aim: to assess the association between two categorical variables
- Condition:
 - Independent samples
 - Variables are categorical
- Example:
 - To determine the association between smoking and exercise
- Assumptions:
 - Expected count < 5 :
 - $>20\%$ - Fisher exact test
 - $<20\%$ - Pearson chi square test

- Presenting and interpreting the result:
 - In sentence: there is no significant association between gender and vaping ($P = 0.140$)
 - In table:

Variable		Lung cancer n(%)	Non-lung cancer n(%)	n	P-value*
Gender	Female	5 (25.0)	15 (75.0)	20	0.140
	Male	0 (0.0)	10 (100.0)	10	

*Fisher's exact test

McNemar's test

- Aim: to assess the association between two categorical variables
- Condition:
 - Dependent/related samples:
 - Matched sample
 - Cross over design
 - Pre and post
 - Variables are categorical
- Example:
 - To determine the impact of an intervention on smoking

- Presenting and interpreting the result:
 - In sentence:
 - There is a significant association between counselling and self-esteem ($P = 0.005$)
 - Proportion of low self-esteem students significantly reduced after the counselling

- In table:

Variable		Post counselling		n	X ² -statistics* (df)	P-value*
		Low self-esteem n(%)	High self-esteem n(%)			
Pre counselling	Low self-esteem	14 (63.6)	8 (36.4)	22	8 (1)	0.005
	High self-esteem	0 (0.0)	3 (100.0)	3		

*McNemar's test

Data - chisquare_smoking.csv

- The data was about 106 patients in Hospital A
- The study aims to determine the association between smoking and lung cancer
- Variables:
 - Id
 - Smoking:
 - 0 = non-smoker
 - 1 = smoker
 - Lung_cancer
 - 0 = non-cancer
 - 1 = cancer



Data - fisher_vaping.csv

- The data was about 30 students in school XYZ
- The study aims to determine the association between gender and vaping
- Variables:
 - Id
 - Gender:
 - 1 = female
 - 2 = male
 - Vaping
 - 0 = no
 - 1 = yes



Data - mcnemar_self_esteem.csv

- The data was about 25 students in school XYZ
- The study aimed to determine the effect of counselling on their self-esteem
- Variables:
 - Id
 - Pre: before counselling
 - 0 = low self-esteem
 - 1 = high self-esteem
 - Post: after counselling
 - 0 = low self-esteem
 - 1 = high self-esteem





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